Kunj Patel

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Education

Boston University, MA, USA | M.S. Applied Data Analytics | GPA: 3.8/4.0 — Sept '25 – Dec '26

Gujarat Technological University, India | B.E. Information Technology | GPA: 3.7/4.0 — Jul '20 – Jun '24

Publications

WikiFlow: Predicting Wikipedia Editor Dropout at Scale Using Distributed Machine Learning on Google Cloud — **CSECS 2026**

K. Patel, A. Meena, F. Alizadeh-Sabdiz, EAI Conference on Computer Science & Education in Computer Science (CSECS 2026), Accepted; presenting Nov 2026

- Jointly built a distributed machine learning pipeline with Apache Spark on Google Cloud (PySpark, Dataproc, GCS) that processed 25GB of MediaWiki history (60M+ revision events) for 1.15M+ editors, 10x larger than any prior single machine study of this problem.
- Engineered 13 behavioral features with no target leakage and built three models from scratch: logistic regression on Spark RDDs (AUC 0.845), clustering of editors into behavioral archetypes, and a distributed deep neural network with Elephas (AUC 0.909, 6.4 points above baseline).
- Documented the pipeline and methodology, delivering the first reproducible benchmark for editor dropout prediction at this scale.

Technical Skills

Languages: Python, SQL, R, Java, JavaScript, TypeScript, LaTeX
Analytics & ML: Machine Learning, Neural Networks, Bayesian Inference, Hypothesis Testing, Statistical Modeling, Exploratory Data Analysis, Feature Engineering, Classification, Regression, Model Deployment, RAG, Scikit-learn, TensorFlow, XGBoost, PySpark, LangChain, FAISS, Pandas, NumPy
Big Data & Databases: Apache Spark, Hadoop, ETL Pipelines, Data Warehousing, Data Validation, PostgreSQL, MySQL, MongoDB
BI & Visualization: Power BI, Tableau, Excel, Streamlit, Jupyter
Cloud & Tools: GCP (BigQuery, Dataproc, GCS), AWS (S3, EC2), Docker, Git, FastAPI, Agile/Scrum

Professional Experience

Data Analyst, Boston University IT Help Center | *Power BI* — May '26 – Present

- Analyzed weekly support call data, segmenting handle time by representative and ticket volume by requester age group, to find recurring bottlenecks and share them with leadership.
- Built weekly and monthly Power BI dashboards, adopted by IT Help Center leadership, that tell the story of resolution trends and team performance.
- Prototyped an AI triage layer that resolves routine tickets and confirms resolution automatically, while leading the team in call volume three weeks in a row (avg. 90 calls/week).

Software Developer, Artem HealthTech Pvt. Ltd. | *Python, SQL, Redis* — May '24 – Sep '24

- Maintained a production Health Management Information System used by 1,000+ municipal healthcare workers; optimized SQL queries to cut page load times 35% and speed up task completion 20%.
- Redesigned 8 core clinical workflows, reducing average navigation from 12 to 7 clicks per task and saving clinicians 45 minutes a day; added a Redis caching layer that lowered database load 40% and cut infrastructure costs by $3,000 per month.

Project Manager, Wellnest Tech | *Angular, .NET, CI/CD* — Jan '24 – May '24

- Designed 8 executive dashboards tracking sales velocity and inventory turnover that identified $50K in recoverable revenue, giving leadership a clear view of financial and operational performance.
- Built a sales analytics platform for 200+ daily users that automated 12 manual reporting processes; promoted after 3 months to lead 5 developers (95% of features delivered on schedule) and cut deployment time from 3 days to 4 hours with CI/CD.

Software Developer Intern, Telnet Pvt. Ltd. | *Angular* — Jul '23 – Aug '23

- Shipped single click add to cart for an online shopping experience, reaching 85% user satisfaction and 70% feature adoption within two weeks of launch.
- Developed a recipe management platform in Angular that let 500+ beta users build shopping lists 30% faster by automatically combining ingredients across recipes.

Project Experience

Healthcare Document Intelligence System | *Python, FastAPI, RAG, FAISS* — Sep '25 – Dec '25

- Designed a RAG pipeline with FAISS that returns semantic search results across 10,000+ documents in under 2 seconds, and built entity extraction for 1,000+ medical records a month (92% F1), cutting manual review time 70%.
- Deployed the models behind a FastAPI backend handling 50 concurrent uploads and files up to 500MB at under a 3% error rate, raising data accuracy from 78% to 95% and saving clinicians 15 hours a week.

Disability Status Prediction on the American Housing Survey | *Python, scikit-learn, XGBoost, imbalanced-learn* — Spring '26

- With one teammate, explored 145 raw survey variables, engineered 778 features, and benchmarked 9 classification algorithms across 5 class balancing strategies (36 model combinations) to predict disability status for 9,241 U.S. households.
- Chose Random Undersampling with tuned Gradient Boosting (ROC AUC 0.842, MCC 0.465) and showed why SMOTE and ADASYN underperformed: interpolating encoded categorical features created household profiles that cannot exist.

Apache Cassandra vs. MongoDB: Distributed Database Benchmark | *Python, GCP, Cassandra, MongoDB* — Spring '26

- Benchmarked both databases on a GCP VM with 1M NYC Taxi rows across four workloads; Cassandra was 2.2x faster on sequential writes, while MongoDB was 5x faster on aggregations and 30x faster on bulk deletes.
- Caught a flaw in the test design (synchronous writes compared against batched writes) that had produced a misleading 120x gap, corrected it, and brought results in line with published research.